

Yuemin(Austin) Yu

📍 Hong Kong/Shanghai ✉ yuyuemin@connect.hku.hk ☎ +852 5575 2418 🌐 yuyueminaustin.github.io
in Yuemin Yu 🌐 YuyueminAustin

Education

-
- The University of Hong Kong** *Sept. 2021 – June 2025*
BEng in Computer Science, Department of Computer Science, School of Computing and Data Science
◦ GPA: 3.56/4.3
- University of Toronto** *Jan. 2024 – Apr. 2024*
Exchange Student, Department of Computer Science, Faculty of Arts & Science
◦ GPA: 3.43/4.0

Projects

-
- Real-Time Respiration Monitoring and Presence Detection**
HKU AIoT Lab
◦ A web application which implemented novel acoustic sensing algorithm
◦ Provided ubiquitous and non-invasive sensing on breathing and presence based on acoustic signals
- IR-UWB Radar based Breathing Rate Estimation**
HKU AIoT Lab, in progress
◦ Applied algorithms on IR-UWB signals to estimate breathing rate of the user
- Geometry Aware Room Impulse Response Generation and Visualization**
HKU AIoT Lab, in progress
◦ Implementing Diffusion-based deep learning models to generate Room Impulse Response
◦ Using React to build a frontend for visualization of the generated signals

Publications

-
- Statistical Acoustic Sensing For Real-Time Respiration Monitoring and Presence Detection**
Sheng Lyu, Ruiming Huang, **Yuemin Yu**, Chenshu Wu
Mobisys' 2024 Demos, DOI: [10.1145/3643832.3661846](https://doi.org/10.1145/3643832.3661846) [🔗](#)

Experience

-
- Research Assistant, [HKU AIoT Lab](#)** *June 2023 - present*
Supervisor: Prof. Chenshu Wu, Department of Computer Science, HKU Hong Kong
◦ Completed Summer Research Internship of the Department, 58 participants in total
- Mitacs Research Intern, [The Hu Lab](#), [Western University](#)** *June 2024 - Sept. 2024*
Supervisor: Prof. Pingzhao Hu, Department of Biochemistry, Western University London, Canada
◦ Worked as Research intern in the lab, with the support of Mitacs Resarch Internship Award
◦ Initiated the project of Machine-learning Assisted Wearable Sensor design
- Student Teaching Assistant, *Department of Computer Science, HKU*** *Sept. 2022 - Nov. 2022*
Hong Kong
◦ Worked as a student teaching assistant for the course ENGG1330, an introductory programming course
◦ Conducted tutorials on a weekly basis, responded to students' questions

Awards

HKU Foundation entrance scholarship, HKU (full tuition)	<i>Sept. 2021 - present</i>
Dean's Award, Faculty of Engineering, HKU	<i>Sept. 2021 - June 2022</i>
Mitacs Globalink Research Internship Award Candidate (Western University)	<i>Feb. 2024</i>
Exchange Scholarship, HKU	<i>Mar. 2024</i>

Abilities

Programming Languages:

Python (Numpy, Pandas, Pytorch), Java, C++, Haskell, HTML&CSS, Javascript, SQL, R

CS Related Courses:

- Core: Computer Organization (A), Discrete Mathematics (A-), Computer Networks (A+), Algorithm Design & Analysis & Complexity (A-), Principles of Operating Systems (A)
- AI related: Introduction to Deep learning for computer vision (A+), Machine Learning (A-)
- Other: Bioinformatics (A), Web Programming (A+), Programming I (Python, A-), Programming II (Linux and C++, A),

Maths & Stats Related Courses:

Fundamental calculus (A+), Linear Algebra & Probability & Statistics (A), Statistics: Idea and Concepts (A), Multivariable Calculus

Speaking Languages:

English (Obtained IELTS 7.5 with 7 in speaking in 2023.03), Mandarin (Native), Cantonese (Basic)

Activities

Peer Mentor of Faculty of Engineering, HKU	<i>Aug. 2023 - present</i>
Member, Team INVENEER, finalist, AWS Educate Business Innovation Competition 2022, AWS HK	<i>Oct. 2022</i>